GAGANDEEP SINGH

Kanpur, India | 9555504621 | gagandeep455789@gmail.com | LinkedIn | GitHub | Portfolio

AI/ML Intern | Software Development | Deep Learning | Generative AI

B.Tech Information Technology student with hands-on experience in Machine Learning, Deep Learning, Computer Vision, Generative AI, and Retrieval-Augmented Generation (RAG). Built production-ready AI applications using Python, PyTorch, FastAPI, and vector databases. Passionate about implementing research papers into scalable AI systems and solving real-world engineering problems.

EDUCATION

- **Pranveer Singh Institute of Technology (PSIT), Kanpur — B.Tech IT**
  Expected Graduation: 2027 | CGPA: 8.1/10 (till 6th semester)

PROFESSIONAL EXPERIENCE

- **Student Research Associate | Indian Institute of Technology (IIT Kanpur) | Remote | Oct 2025 – Jan 2026**
  - Developed Python-based modules for Virtual Reality and intelligent systems research.
  - Built experimental pipelines for data processing, automation, and technology evaluation.
  - Collaborated with researchers to optimize system performance and document technical findings.
- **Artificial Intelligence Intern | SmartBridge (Google for Developers) | Remote | Feb 2026 – Apr 2026**
  - Built REST APIs and ML pipelines using Python, improving model deployment and integration workflows.
  - Trained and optimized ML models using Python, Scikit-learn, and TensorFlow.
  - Improved model performance through preprocessing, hyperparameter tuning, and evaluation techniques.

PROJECTS

- **BuildWise AI | Generative AI, RAG, Full-Stack Development | GitHub Repository | Live Preview**
  - Built a production-ready AI compliance assistant using FastAPI, Next.js, Supabase, PostgreSQL, and OpenAI APIs.
  - Implemented Retrieval-Augmented Generation (RAG) using vector embeddings, semantic search, metadata filtering, and document chunking for grounded responses.
  - Developed scalable backend APIs with multilingual support, automated report generation, and real-time document understanding.
- **CodeSherpa AI | Generative AI, Repository Intelligence, Full-Stack Development | GitHub Repository | Live Preview**
  - Built an AI-powered GitHub repository assistant using LLMs, semantic retrieval, and vector embeddings.
  - Implemented intelligent codebase search, dependency analysis, and contextual code understanding.
  - Developed interactive architecture visualizations using FastAPI, Next.js, and D3.js.
- **Skin Cancer Classification using Deep Learning | TensorFlow, Computer Vision, Transfer Learning | GitHub Repository**
  - Built an automated skin cancer classification pipeline using Computer Vision, CNNs, Vision Transformers, and transfer learning.
  - Trained and benchmarked multiple deep learning models with advanced optimization techniques including fine-tuning, CBAM attention, early stopping, and data augmentation.
  - Achieved 97.7% binary classification accuracy and 92% multi-class accuracy across 14 skin lesion categories.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, SQL, JavaScript
- **AI/ML:** Machine Learning, Deep Learning, Generative AI, LLMs, RAG, LangChain, NLP
- **Frameworks:** FastAPI, Flask, PyTorch, TensorFlow, Scikit-learn, React.js, Next.js
- **Development:** REST APIs, Backend Development, Full-Stack Development
- **Databases:** PostgreSQL, MySQL, MongoDB, FAISS, Supabase
- **Cloud & Tools:** OCI, AWS (Basic), Docker, Git, Jupyter Notebook, VS Code
- **Core CS:** DSA, OOP, DBMS, System Design

CERTIFICATIONS

- **Machine Learning Specialization — Stanford University (Andrew Ng) | 2025:** Focused on supervised & unsupervised learning, model evaluation, and optimization techniques.
- **Generative AI with Large Language Models** — **DeepLearning.AI & Amazon Web Services | 2026:** Covered LLMs, Retrieval-Augmented Generation (RAG), fine-tuning, and Generative AI applications.
- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle | 2025:** Developed and deployed Generative AI solutions using OCI services and AI infrastructure.
- **Amazon ML Summer School 2025 — Amazon | 2025:** Covered machine learning fundamentals, model development, evaluation, and predictive analytics.

KEY ACHIEVEMENTS & ADDITIONAL HIGHLIGHTS

- Winner – Secured 1st place for developing a scalable Virtual Lab solution**,** demonstrating strong system design and execution.
- Co-inventor of two Indian patents — *Adaptive DNA Sequence Analysis Using BLT* & *Hybrid Deepfake Detection System*.
- Tech Expo Recognition **–** Developed an innovative life-saving steering system for real-time embedded safety.
- Exhibitor **– India AI Impact Summit:** Showcased the Virtual Lab platform to industry professionals and AI practitioners.
- Published 2 peer-reviewed international research papers in Computer Vision, Deep Learning, and AI Optimization.
- Presented research at **EACE 2025 & ICNGTSE 2025**; participated in **Engineers Conclave (IIT Kanpur)**.